Newton's 3rd Law Worksheet Name _____

Choose the best answer for each question from the choices below then explain why you have chosen the answer you chose. Good Luck!!!

1. Newton's 3rd Law states…
   a. Objects in motion stay in motion and objects at rest stay at rest
   b. Force is equal to mass times acceleration
   c. For each action there is an equal and opposite reaction

Why???_____

2. An archer shoots an arrow. The action force is the bowstring against the arrow, The reaction force is…
   a. Air resistance against the bow
   b. Arrow's push against the bowstring
   c. Grip of the archer's hand on the bow

Why???_____

3. A player catches a ball. The action force is the impact of the ball against the player's glove. The reaction force is…
   a. The force the glove exerts on the ball
   b. The player's grip on the glove
   c. The friction of the ground on the player's shoes

Why???_____

4. A player hits a ball with a bat. The action force is the impact of the bat against the ball. The reaction force is…
   a. The grip of the player's hands on the ball
   b. The air resistance on the ball
   c. The force of the ball against the bat

Why???_____

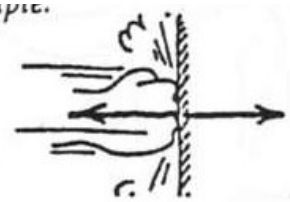
5. A baseball player bats a ball with a force of 1,000 N. The ball exerts a reaction force against the bat of…
   a. Less than 1,000 N
   b. More than 1,000 N
   c. 1,000 N

Why???_____

6. A person is attracted toward the center of the Earth by a 500 N gravitational force. The force that the Earth is attracted toward the person is…
   a. 500 N
   b. Much less than 500 N
   c. Much more than 500 N

Why???_____

Example:



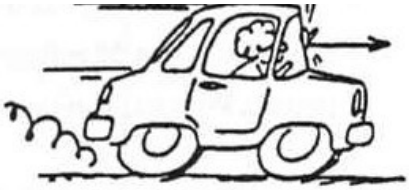
Fist hits wall.

Wall hits fist.



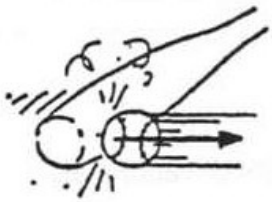
Head bumps ball.

(a)_____



Windshield hits bug.

(b)_____



Bat hits ball.

(c)_____



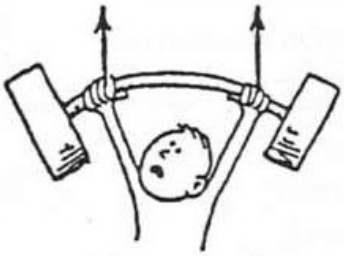
Hand touches nose.

(d)_____



Hand pulls on flower.

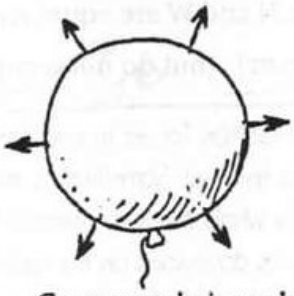
(e)_____



Athlete pushes bar upward.

(f)_____

_____



Compressed air pushes balloon surface outward.

(g)_____

_____

(h)_____

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2. Draw arrows to show the chain of at least six pairs of action-reaction forces below.

